WORKED SOLUTION

STEP I 1999, Question 3

Pure • Cyclic Systems of Equations

STEP I 1999, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q3

The key idea

Each equation says $x_k = f(x_{k+1})$ with $f(x) = 1 + 1/x$, and the system closes up into a loop. Differences between neighbours obey a simple rule: each difference is the next one divided by a product of two numbers bigger than 1. So going round the loop the differences can only shrink. Since we come back to where we started, they must all be zero.

Throughout, read the indices cyclically: x_{n+1} means x_1 and x_{n+2} means x_2. Then every equation in the system has the single form

$$x_k = 1 + \frac{1}{x_{k+1}}, \qquad k = 1, 2, \ldots, n.$$

Solution

Part (i)

All the x_k are positive, so $1/x_{k+1} > 0$ for every k. Hence

$$x_k = 1 + \frac{1}{x_{k+1}} > 1 \qquad \text{for } k = 1, 2, \ldots, n,$$

and because of the cyclic indexing this covers every one of $x_1, \ldots, x_n$ (the last one is $x_n = 1 + 1/x_1$). So $x_1, x_2, \ldots, x_n > 1$.

Part (ii)

Subtract the second equation from the first:

$$x_1 - x_2 = \left(1 + \frac{1}{x_2}\right) - \left(1 + \frac{1}{x_3}\right) = \frac{1}{x_2} - \frac{1}{x_3} = \frac{x_3 - x_2}{x_2 x_3} = -\frac{x_2 - x_3}{x_2 x_3},$$

which is the required result. We used only $x_1 = 1 + 1/x_2$ and $x_2 = 1 + 1/x_3$ (and $n \geq 3$, so that x_3 is a different variable). Exactly the same calculation for any two consecutive

equations gives, for every k,

$$x_k - x_{k+1} = -\frac{x_{k+1} - x_{k+2}}{x_{k+1}x_{k+2}} \qquad \text{(indices cyclic)}.$$

Part (iii)

Write $d_k = x_k - x_{k+1}$ for the differences between neighbours, so $d_n = x_n - x_1$. The general form of (ii) reads $d_k = -d_{k+1}/(x_{k+1}x_{k+2})$. Taking absolute values, and using part (i), which gives $x_{k+1}x_{k+2} > 1$,

$$|d_k| = \frac{|d_{k+1}|}{x_{k+1}x_{k+2}} \leq |d_{k+1}|,$$

with *strict* inequality whenever $d_{k+1} \neq 0$. Chaining these round the loop,

$$|d_1| \leq |d_2| \leq |d_3| \leq \cdots \leq |d_n| \leq |d_1|.$$

The two ends are the same, so every inequality in the chain is an equality. In particular $|d_1| = |d_2|$. But if $d_2 \neq 0$ we would have $|d_1| < |d_2|$ strictly, so $d_2 = 0$, and then all the $|d_k|$, being equal to $|d_2|$, are zero.

Ask yourself Where exactly did we use that the system closes up, with $x_n = 1 + 1/x_1$?

Only in the last link of the chain, $|d_n| \leq |d_1|$, which comes from $x_n = 1 + 1/x_1$. Without it the differences could simply grow as k increases, and indeed an open chain $x_1, \ldots, x_n$ need not be constant. It is the return to the start that forces the shrinking to be no shrinking at all.

So $x_k = x_{k+1}$ for every k, that is, $x_1 = x_2 = \cdots = x_n$.

The value of x_1

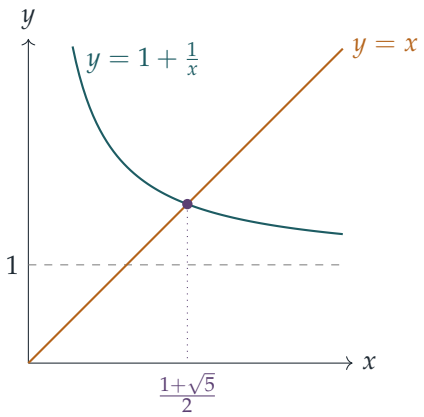
Call the common value x. Any one of the equations now says $x = 1 + 1/x$, so

$$x^2 - x - 1 = 0, \qquad x = \frac{1 \pm \sqrt{5}}{2}.$$

The negative root is ruled out because $x > 1$ by part (i), so

$$x_1 = \frac{1 + \sqrt{5}}{2}.$$

Conversely, setting every x_k equal to this value does satisfy all n equations, so the system really has this solution, and by (iii) it has no other. Geometrically, x is where the curve $y = 1 + 1/x$ meets the line $y = x$:



Checks

- With $x = \frac{1+\sqrt{5}}{2} \approx 1.6180$ we get $1 + 1/x \approx 1 + 0.6180 = 1.6180$, as required.
- The identity in (ii) can be tested on any open chain: take $x_3 = 2$, so $x_2 = \frac{3}{2}$ and $x_1 = \frac{5}{3}$. Then $x_1 - x_2 = \frac{1}{6}$ and $-(x_2 - x_3)/(x_2 x_3) = \frac{1/2}{3} = \frac{1}{6}$.
- Going all the way round the loop, $x_1 = f^n(x_1)$ with $f(x) = 1 + 1/x$. A numerical scan of $f^n(x) - x$ on $(0, 20)$ for $n = 3, \ldots, 11$ finds a single sign change, at $x \approx 1.618$, confirming that there is no other solution.

Common slips

- Proving (iii) from $|d_1| \leq |d_2|$ alone. One step only says the differences don't grow; you need to go all the way round the loop and come back to $|d_1|$.
- Assuming the x_k are equal and then finding x_1. Part (iii) is what justifies that, and it needs the strict inequality $x_{k+1}x_{k+2} > 1$ from part (i).
- Forgetting to reject $\frac{1-\sqrt{5}}{2}$, which is negative.

Beyond STEP

The heart of this question is that $f(x) = 1 + 1/x$ brings points closer together on $x > 1$, since $|f(a) - f(b)| = |a - b|/(ab)$. A map like this is a *contraction*, and the Banach fixed-point theorem says that a contraction of a complete space has exactly one fixed point, which iteration from any start converges to. Iterating here from $x = 1$ gives the ratios of consecutive Fibonacci numbers, $1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \ldots$, and the continued fraction $1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}$ for the golden ratio. For the analysis, read Abbott, *Understanding Analysis*. For continued fractions, read Davenport, *The Higher Arithmetic*.

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